Personalised Reward Models Learn Little About Which Response a New Rater Prefers: A Systematic Test with a Scale-Use Baseline

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Abstract

Personalised and pluralistic reward models promise to represent how individual people judge model outputs — through demographic groups, learned user latents, or examples placed in context. We test what such models actually learn about raters they have never seen and find that none of them predicts better *which* response a new rater prefers; what they can learn is *how strongly* the rater expresses preferences. We test this under one protocol on three public multi-rater datasets (MultiPref pairwise preferences, DICES conversational-safety ratings, PRISM response ratings): new raters, k of their own ratings for adaptation, evaluation on held-out items. As a reference we use a deliberately simple baseline from measurement theory: a few text-independent offsets per rater on a distributional reward model's output, describing how the rater uses the scale (three numbers on two-sided preferences: how often they say "clearly", "slightly", "tie"). A learner-free analysis first shows where the predictable structure sits: beyond item location, rater-side scale use improves held-out likelihood about 25 times more than item-specific dispersion, while the item-level shape that replicates across rater panels is small and only partly decodable from text. Under the common protocol the scale-use baseline improves held-out log-likelihood by 0.08–0.10 nats per rating at k = 20 on all three datasets; demographic or crowd/expert groups add at most 0.003; a tuned low-rank text-interacting latent matches but does not exceed it; reimplementations of two published personalised reward models (VPL, PAL) gain less, and their user representations do not encode scale use; an in-context-personalised LoRA reward model gains less (five folds; also without its one failed fold); and the gain is unchanged on top of a fine-tuned reward model. Two properties limit what this personalisation buys: on binary preference labels — the format most personalisation methods are trained and evaluated on — every method's gain is at most 0.002 nats, and the gain is mostly calibration (how strongly a rater prefers) rather than ranking (which response they prefer; +0.5 points of direction accuracy on MultiPref). Across group-conditioned, low-rank, published (VPL, PAL), in-context and fine-tuned personalisation, no method improves new-rater direction accuracy over its own unpersonalised version by more than a point on pairwise preferences, and on 41,199 native pairwise choices of PRISM users every method's improvement in a new user's choice accuracy has an upper 95 % bound below one point (fixed before the result). We recommend graded labels, new-rater evaluation, separate calibration and ranking metrics, and a scale-use calibration baseline for personalised reward modelling.

1. Introduction

Reward models and LLM judges are trained on human ratings that disagree. A recent line of work treats this disagreement as signal to be modelled rather than noise to be averaged: distributional preference learning estimates a distribution over each alternative's utility rather than a point score (Siththaranjan et al., 2024); jury learning predicts individual annotators from content, annotator and demographic-group embeddings (Gordon et al., 2022), and large datasets such as PRISM are analysed by sociodemographic group (Kirk et al., 2024); personalised reward models learn a per-user latent, inferred for a new user from a few examples (Poddar et al., 2024; Chen et al., 2025). The pluralistic-alignment agenda (Sorensen et al., 2024) makes the representation of individual and group differences a first-class goal.

What these models learn about a person is rarely tested directly. Evaluations typically use synthetic users whose preferences differ by construction, binary preference labels,

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or users seen in training. We ask a simple question under a common protocol on real raters: **given k ratings from someone the model has never seen, how much better can it predict their other ratings — in particular, which response they prefer — and what is it that it learns?** The answer we find is that the methods learn calibration, not ranking.

Measurement theory suggests a candidate answer that reward modelling largely ignores: raters differ in **how they use the response scale** — some say "clearly better" where others say "slightly better", some rarely call a tie, some are harsher across the board (Linacre, 1989; Baumgartner & Steenkamp, 2001). Such response styles are text-independent, cheap to estimate, and invisible to models that condition only on content or demographics. We make them a reference point for calibration. Because they are text-independent they cannot, by construction, change which response a rater prefers; a personalised reward model that learns content preferences should therefore show its value in ranking, and a model whose likelihood gain does not exceed the scale-use offsets has not shown that it learns more than scale use.

Contributions. (1) A learner-free measurement of where the predictable structure of multi-rater preference data sits — split-half reliability of the rating distribution's shape, cross-fitted text decodability, and a rater–item decomposition of held-out likelihood (§3). (2) A three-parameter scale-use baseline that plugs into any reward model with graded outputs (§4). (3) A systematic new-rater test, on three datasets, of group-conditioned, low-rank-latent, published (VPL, PAL), in-context and fine-tuned personalised reward models against that baseline, with hypotheses fixed in time-stamped internal protocols, post-result changes marked as amendments, and failures reported (§5, §8). (4) Two limits of what personalisation buys — it disappears on binary labels and it is mostly calibration rather than ranking — and recommendations for the field (§6–7).

2. Related work

Pluralistic and distributional reward models. Distributional preference learning (Siththaranjan et al., 2024), diverging-preference detection (Zhang et al., 2025), PRISM (Kirk et al., 2024), MultiPref (Miranda et al., 2025) and DICES (Aroyo et al., 2023) motivate modelling disagreement; jury learning (Gordon et al., 2022) and multi-annotator models (Mostafazadeh Davani et al., 2022) predict individual annotators. **Personalised reward models** infer a user latent from a few examples (Poddar et al., 2024; Chen et al., 2025), represent users in a low-rank reward subspace (Bose et al., 2025), induce persona prompts from a user's history (Ryan et al., 2025, who also release a benchmark of PRISM and Chatbot Arena users), or learn per-user embeddings jointly with the reward model (Li et al., 2024); VPL's latent can collapse under sparse preference data (Kim & Kim, 2026). A multi-faceted comparison of personalisation methods (Dong et al., 2025) reports large differences between methods and safety costs. Annotator embeddings (Deng et al., 2023) and per-annotator heads (Mostafazadeh Davani et al., 2022) are close relatives of our offsets; our contribution is not the offset model but the new-rater test and what it shows. Rater groups have been found to use safety severity scales differently (Mishra et al., 2026), which is consistent with the leniency component we find on DICES. **Annotator demographics** as predictors of labels have been found weak in several studies (e.g. Orlikowski et al., 2023). **Measurement theory**: rater severity and scale use are standard facets in many-facet Rasch measurement (Linacre, 1989; Eckes, 2011) and response styles are a classical survey-methods topic (Baumgartner & Steenkamp, 2001). We connect the last line to the first: for new-rater prediction, the measurement-theory rater facet is what the reward model is missing.

3. Where is the predictable structure? A learner-free measurement

Data. MultiPref (Miranda et al., 2025; 10,461 comparisons; 4 ratings each by 2 crowd and 2 expert raters; 227 persistent raters; 5 ordered levels A-clearly … B-clearly). PRISM (Kirk et al., 2024; 68,371 responses each rated 1–100 by one of 1,396 users; binned to 5 fixed levels). DICES-350 (Aroyo et al., 2023; 350 conversations fully crossed with 104 quality-checked raters; safe / unsure / unsafe).

Estimators. For an item with ratings on K ordered levels, let h be the empirical rating distribution and C the cumulative operator. Split the item's raters at random into disjoint panels P_1 (a raters) and P_2, and let q be a cross-fitted *location-only* forecast with exactly the mode and mean of P_1 (built from other items in the same mode-and-mean cell). With $r_1 = C(h_1 - q)$, $r_2 = C(h_2 - q)$, $M = \|r_1\|^2/n$ and $X = \langle r_1, r_2\rangle/n$, the forecasting value of the raw distribution over its own location obeys the identity $S(h_1) - S(q) = M - 2X$ under the ranked probability score, so $\rho = X/M$ is a split-half reliability of the distribution's *shape* and the raw distribution beats its location exactly when $\rho > 1/2$. Text decodability is the proper-score gain $G = (2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2)/n$ of a residual $\hat{r}$ decoded from frozen text features, cross-fitted over prompt groups, reported as the fraction G/X of the reproducible shape. The rater–item decomposition fits ordered-probit models $Y^*_{ir} = \theta_i + \delta_r + \sigma_i\sigma_r\varepsilon$ with item locations θ_i, rater shifts δ_r, rater scales σ_r and item dispersions σ_i, compares held-out likelihood over random cells (every held-out rater and item appears in training), and chooses all penalties by inner cross-validation.

Reproducibility. With the split-half estimator of a companion study (anonymised), the reliability of the rating distribution's shape beyond its exact mode and mean is $\rho = 0.25$ [0.22, 0.29] on MultiPref overall preference with three raters (helpful 0.29, harmless 0.21): the raw few-rater distribution forecasts a new rater worse than its own location ($\rho < \frac{1}{2}$). **Text decodability.** A cross-fitted ridge decoder on frozen Qwen2.5-7B-Instruct features, scored on an independent rater panel, recovers G/X = 10.5 % [5.7 %, 11.9 %] of the reproducible shape (within-cell permutation control ≤ 0; positive control 0.35) — comparable to the upper end for ordinal quality ratings (upper bounds 0.4–10 %) and below NLI (17–26 %). **Rater vs item.** A long-format ordered-probit decomposition with real rater identities and penalties tuned by inner cross-validation: rater-specific thresholds improve held-out likelihood by +0.036 nats per rating over item location alone, item-specific dispersion by a further +0.0014 (ratio 25; 19–36 across the five folds); of the rater part, rater scale contributes +0.031 and rater shift +0.005.

Beyond item location, the predictable structure is therefore mostly rater-side scale use, while the item-level shape that replicates across disjoint panels is small and only partly in the text. These two facts come from different estimands and do not say that disagreement is "not in the text". The stage-1 decision rule we had written down returned **"mixed"**, because G/X was not ≤ 10 % (interval 5.7–11.9 %); the factorised model of §4 was chosen with that outcome known.

4. A three-parameter scale-use baseline

(Figure 2 illustrates the effect of the offsets.) Let $\ell(x) \in R\hat{}K$ be the class logits of a text-only distributional reward model for item x (K ordered rating levels). For rater r we add a text-independent offset vector: p(y | x, r) = softmax(ℓ(x) + b_r). Variants: **DF** (b_r $\in R\hat{}K$, free), **DFs** (two-sided preference scales only: b_r tied symmetrically, (e, s, t, s, e) — extreme, slight, tie — 3 parameters, no A-vs-B position term). Rater offsets have a Gaussian prior centred on the training-rater mean; a new rater's offsets are the MAP estimate from k of their ratings with the text model frozen. Baselines under the same protocol: **D** (text-only distributional RM), **DG** (group offsets: crowd/expert, race, region), **DL** (low-rank text-interacting latent: ℓ(x) + (Wx + w_0)·z_r, d = 8, the personalised-RM family). An interpretable probit family (rater shift / log-scale on an ordered-probit text head) is reported as a secondary analysis.

Training and adaptation. The text model and the training raters' offsets are fitted jointly by penalised maximum likelihood,

$$\min \sum_{(i,r,y)} -\log p(y \mid x_i, r) + \lambda_{\text{text}} \|V\|^2 + \lambda_r \sum_r \|b_r\|^2,$$

with λ_{text} chosen on the text-only model and λ_r on held-out items of the training raters. A new rater r^* is adapted from k ratings $\{(x_j, y_j)\}$ by $\hat{b}_{r^*} = \arg\min_b \sum_j -\log \text{softmax}(\ell(x_j)+b)_{y_j} + \lambda_r \|b-\bar{b}\|^2$, with $\bar{b}$ the training-rater mean (the prior mean; $\hat{b} = \bar{b}$ at $k = 0$) and the text model frozen — a convex problem in 3–5 parameters, solved in milliseconds. Any text model that outputs graded log-probabilities can be used: we use a ridge head on frozen LLM features (§5.1–5.3), a LoRA-fine-tuned LLM reward model (§5.4) and the decoder of a published personalised reward model (§5.5).

5. Experiments

Protocol. 5 folds; in each, 20 % of raters (stratified by group) and 20 % of prompt groups are held out; models are trained on training raters × training items with penalties chosen on a validation split; each test rater is adapted on k ∈ {0, 5, 10, 20, 50} of their ratings of *training* items (5 random draws) and evaluated on their ratings of *held-out* items. Primary metric: held-out log-likelihood per rating; also RPS and preference-direction accuracy; rater-cluster bootstrap (B = 2,000). Because test raters share test items on MultiPref and DICES, the headline k = 20 contrasts were also re-estimated with a two-way bootstrap that resamples raters and prompt groups independently: every log-likelihood interval widens by at most 0.006 on either side and no log-likelihood conclusion changes; the one claim that changes is noted where it appears. Text model in §5.1–5.3: frozen Qwen2.5-7B features with a ridge head; §5.4: a LoRA-fine-tuned reward model.

5.1. Main result (MultiPref)

Figure 1 shows the gain over the text-only reward model as a function of k on all three datasets. In the table, DF/DFs/DG use the training-rater prior and DL and the DFs − DL column use the few-shot-tuned prior (the DFs − DL contrast is post hoc).

Preference-direction accuracy (D: 0.674): DFs +0.005 [+0.001, +0.008] at k = 20 (not significant once items are also resampled: [−0.001, +0.011]); DF −0.014 [−0.020, −0.007]. A three-number style model gains as much as a tuned 8-dimensional text-interacting latent (tie at k = 20) and more at k = 5, 10 and 50; fitting a rater's A-vs-B position from few ratings costs direction accuracy.

Reference points. To size these gains: on MultiPref at k = 20 a uniform forecast scores −1.609 nats per rating, a forecast that ignores the text and uses only the new rater's 20 ratings −1.487, the text-only model −1.441, the text model with the three offsets −1.342, and offsets fitted in-sample on all of the rater's held-out ratings (an optimistic ceiling for text-independent offsets) −1.262. Twenty ratings of a person, without the text, recover 73 % of the text model's

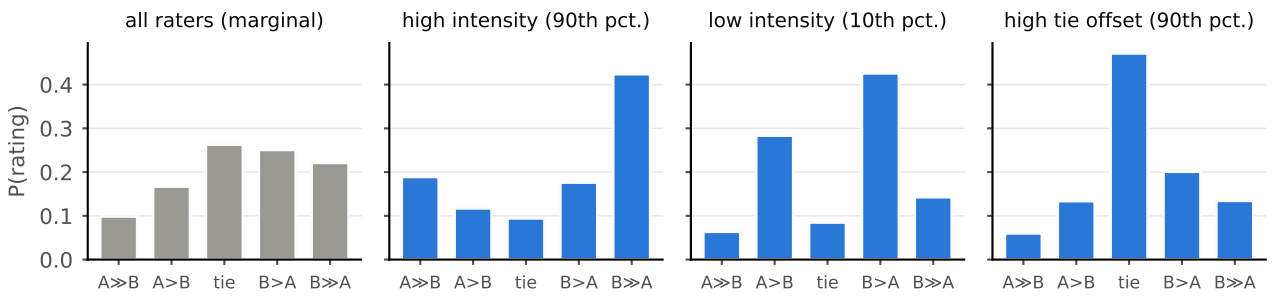


Figure 1. What three numbers per rater change. Left: the marginal distribution of MultiPref ratings. Right: the same distribution passed through the fitted scale-use offsets (extreme, slight, tie) of three real new raters (k = 50) closest to the 90th / 10th percentile of intensity (extreme minus slight) among raters whose tie offset is at or below the median, and closest to the 90th percentile of the tie offset. The offsets are text-independent: they reshape how strongly a rater is predicted to prefer, not which side.

k	DF − D	DFs − D	DG − D	DL − D (tuned prior)	DFs − DL (tuned, post hoc)
5	+0.028 [+0.012, +0.044]	+0.048 [+0.029, +0.070]	≈ 0	+0.023 [−0.002, +0.047]	+0.026 [+0.016, +0.037]
20	+0.080 [+0.055, +0.106]	**+0.100 [+0.074, +0.128]**	≈ 0	+0.095 [+0.069, +0.122]	+0.0045 [−0.0004, +0.0092]
50	+0.114 [+0.087, +0.142]	+0.128 [+0.101, +0.156]	≈ 0	+0.120	+0.007 [+0.002, +0.013]

gain over uniform; the offsets close 55 % of the gap between the text-only model and the in-sample ceiling. On DICES the rater's ratings alone predict better than the text alone (−0.766 vs −0.840 at k = 20). Even the in-sample ceiling raises MultiPref direction accuracy by only 1.3 points (0.674 → 0.687), confirming that text-independent offsets cannot carry ranking.

5.2. Replications (k = 20)

DF − D and DG − D use the training-rater prior, DF − DL the few-shot-tuned prior. Direction accuracy on PRISM means good-vs-bad around the scale midpoint.

5.3. What the offsets encode

In the probit family, rater *scale* matters more than rater *shift* on MultiPref (+0.036 [+0.020, +0.057] at k = 20), but rater *shift* dominates on DICES (−0.053) and PRISM (−0.031): leniency/strictness on single-response rating scales, intensity on two-sided preference scales.

5.4. On a fine-tuned reward model

With a LoRA-fine-tuned Qwen2.5-7B reward model (D-LoRA, trained per fold on the training raters' rating distributions) as the text model, the three-number style model still adds +0.100 [+0.071, +0.130] nats per rating at k = 20 — the same as its gain on the frozen-feature head (+0.100). Fine-tuning improves the text-only model by +0.027 [+0.016, +0.039] (direction accuracy +1.4 points) and the style model by the same +0.027: the two gains add. Personalising the frozen-feature model with 20 ratings per rater gains more than fine-tuning the text model (+0.072 [+0.041, +0.106]). DFs keeps the direction accuracy of D-LoRA (+0.002 [−0.002, +0.006]). Against in-context personalisation — a separate LoRA model on the same base, trained with ten of the rater's other ratings in the prompt and evaluated with the new rater's k ratings in the prompt (one training seed per fold) — the three offsets on D-LoRA are better at k = 20 by +0.062 [+0.043, +0.081] nats per rating over all five folds (209 raters) and by +0.026 [+0.010, +0.043] when the one fold in which the in-context model trained badly is excluded (its gain over D-LoRA by fold: +0.075, +0.021, +0.061, −0.055, +0.059). The in-context result depends strongly on the training seed: retrained with a second seed, the failed fold gains +0.108 instead of −0.055; with that run in place of the first, the offsets are still better by +0.026 [+0.012, +0.041] and still more accurate on direction (+2.4 points [+1.3, +3.4]). Second seeds for the other four folds are running; B3 will be reported for each seed and for their average, and more accurate on which response the rater prefers (+3.2 points [+1.9, +4.4]; +2.9 without that fold). The in-context model does learn from the history in four of five folds, at several thousand prompt tokens per prediction; it never improves direction accuracy over D-LoRA.

5.5. Published personalised reward models

We reimplemented VPL (Poddar et al., 2024) and PAL-B (Chen et al., 2025) from their official code as heads on the same frozen per-response embeddings and evaluated them under the same new-rater protocol, refitting the text-only and style models on those embeddings. On graded

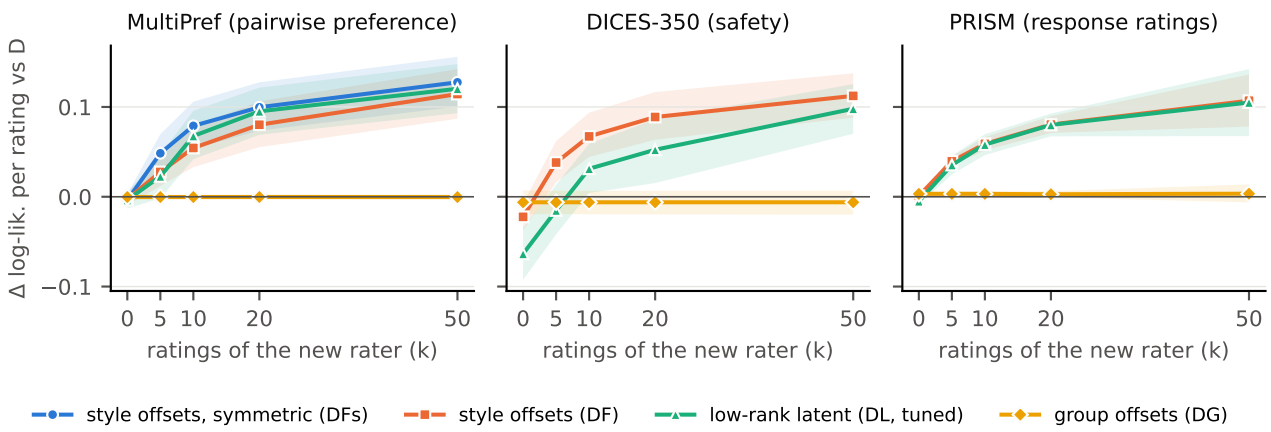


Figure 2. New raters: gain in held-out log-likelihood per rating over the text-only distributional reward model (D), as a function of the number k of the new rater's ratings used for adaptation; 95% rater-cluster bootstrap bands, 5 folds. Text-independent rater offsets (DF; symmetric three-parameter DFs on two-sided preferences) gain 0.08–0.10 nats per rating at k = 20 on all three datasets; group offsets (crowd/expert, race, region) gain nothing; a low-rank text-interacting rater latent with a few-shot-tuned prior matches but does not exceed them (it beats the 5-offset DF on MultiPref and ties DFs). DF, DFs and DG use the training-rater prior; DL the few-shot-tuned prior.

dataset	DF − D	DG − D (groups)	DF − DL (tuned prior)
DICES-350 (safety)	+0.089 [+0.064, +0.117]	−0.006 (race)	+0.037 [+0.022, +0.053]
PRISM (ratings)	+0.080 [+0.071, +0.090]	+0.003 (region)	−0.001 [−0.006, +0.003] (formally not non-inferior: bound −0.006 < −0.005)

labels, at k = 20 the style model is better than VPL by +0.092 [+0.068, +0.119] nats per rating ([+0.063, +0.124] when prompt groups are also resampled) and than PAL-B by +0.158. VPL's latent does carry rater information once its posterior collapse is prevented (KL weight selected as 0 or 0.001; +0.087 over its own unpersonalised version), but only lifts it to the level of the text-only model; PAL-B's few-shot user weights give no gain. Their user representations are unrelated to response style (cross-validated $R^2 \approx 0$ against the style offsets). On binary labels — the setting in which both methods were proposed — every method's personalisation gain is at most 0.002 nats (small but nonzero for PAL-B with K = 4 and for VPL at k = 50), and VPL's text head is the best binary model (the style model is 0.010 nats behind it and 0.005 behind the text-only model, differences of the size of item-sampling noise): the individual differences that are predictable for new raters live in the graded levels and ties that binary preference data discard. VPL's per-response reward head is slightly better at predicting *which* response is preferred (0.677 vs 0.668), an effect present without personalisation. The offsets are also a plug-in: added to VPL (fitted on the same k ratings, VPL frozen) they improve it by +0.088 [+0.066, +0.111] nats per rating at k = 20, tie with the style model on likelihood (−0.004 [−0.014, +0.005]) and keep VPL's direction accuracy (+1.1 points over the style model alone).

5.6. Summary of baselines (MultiPref, new raters, k = 20)

Gains are relative to each block's text-only model (D, Dp, D-LoRA); the LoRA block is restricted to the 209 raters with an in-context evaluation at k = 20 (all five folds).

Rows within a block share the text model and the raters; across blocks the text models differ (text-only direction accuracy 0.674 / 0.670 / 0.688). VPL's and PAL-B's own gains over their unpersonalised versions are +0.087 and −0.000.

6. What personalisation buys: calibration, not ranking

Binary labels remove the signal. On MultiPref with ties dropped and labels binarised, every method's personalisation gain is at most 0.002 nats per rating (§5.5), and the scale-use baseline's binary accuracy equals the text-only model's (−0.0002). The individual differences that are predictable for new raters live in the graded levels and in ties — exactly what binary preference data discard, and the format in which most personalised reward models are trained and evaluated. **Ranking moves little.** Text-independent offsets change how strongly a rater is predicted to prefer a response, not which response: on MultiPref the scale-use baseline raises direction accuracy by only +0.005 [+0.001,

text model	personalisation	gain in LL per rating vs the block's text-only model	direction accur
frozen features (joint item embedding)	group offsets (crowd / expert)	≈ 0	0.677
	low-rank latent, tuned prior	+0.095 [+0.069, +0.122]	0.679
	style offsets, 3 numbers (DFs)	**+0.100 [+0.074, +0.128]**	**0.679**
frozen features (per-response embeddings)	VPL (reimplemented)	+0.002 [−0.008, +0.012]	0.677
	PAL-B (reimplemented)	−0.065 (worse)	0.666
	DFs	**+0.094 [+0.069, +0.120]**	0.668
	VPL + style offsets	+0.090	0.679
LoRA-fine-tuned Qwen2.5-7B (209 raters)	in-context rater history (RC-LoRA)	+0.030 [−0.001, +0.061]	0.658
	DFs	**+0.092 [+0.063, +0.125]**	0.689

+0.008] at k = 20 ([−0.001, +0.011] with items also resampled) (and the 5-offset version with an A-vs-B term lowers it by 0.014), while RPS drops from 0.1593 to 0.1532. On single-response safety ratings the same offsets do move the safe/unsafe decision (+4.8 points on DICES), because there a rater's threshold *is* the decision. Content-interacting personalisation does not do better on ranking: VPL's advantage in direction accuracy is present without personalisation (k = 0) and comes from its text head. Against its own k = 0 version, no method's direction accuracy on MultiPref rises by more than 0.1 points at k = 20, and every upper 95 % bound stays below +0.8 points even when prompt groups are resampled together with raters (VPL +0.07 [−0.57, +0.74], style model −0.10 [−0.69, +0.54], PAL-B −0.48 [−1.11, +0.14]). **The ranking null replicates on real users' own choices.** PRISM users each saw 2–4 responses to their own prompt, scored them and chose one; pairing the chosen response with every other one gives 41,199 native pairwise choices of 1,396 users. Under the same new-user protocol (5 user folds × 5 conversation folds, k of the user's pairs from training conversations), we fixed before any result that the ranking null would be stated only if every method's improvement in choice accuracy over its own k = 0 version had an upper 95 % bound below one point at k = 20. It holds for all of them (1,033 new users): VPL +0.12 points [−0.08, +0.33], PAL-B +0.0 [−0.4, +0.4] (K = 2) and +0.3 [−0.1, +0.7] (K = 4), the 5-offset model −0.6 [−1.4, +0.2]; the symmetric scale-use model is identical to the text-only model on binary choices, because a binary label carries no intensity. Personalisation also leaves the binary log-likelihood unchanged (VPL −0.0002, PAL-B −0.0002 nats). On the same data PAL-B's text head is 2.4 points less accurate than the text-only model and VPL's is level with it (+0.2 [−1.0, +1.4]). With a graded label cut from the users' own score differences (thresholds fixed in advance), the MultiPref pattern returns at half the size: at k = 20 the scale-use model gains +0.043 [+0.031, +0.057] nats per rating over its k = 0 version, VPL +0.015 [+0.005, +0.027], PAL-B 0.000; the scale-use model is ahead of VPL by +0.045 [+0.030, +0.062]; VPL plus the offsets gains +0.060 and ties the scale-use model (−1.4360 vs −1.4354); and no method's own direction-accuracy gain exceeds +0.2 points (the 5-offset model loses 1.7). **Scale use does not corrupt the shared reward model.** Training the text model jointly with rater offsets and then using the population-mean offsets (k = 0) does not give a better consensus model than ordinary training: −0.003 [−0.010, +0.003] nats on MultiPref (direction +0.3 points, n.s.), −0.022 [−0.037, −0.008] on DICES, +0.001 on PRISM, −0.009 [−0.016, −0.004] with the LoRA reward model. Ordinary training averages scale use out; it matters for predicting a person, not for learning the content preference.

7. Recommendations

(i) Collect and keep graded labels (and ties) when individual differences matter; binary preferences erase most of what is predictable about a new rater. (ii) Report a scale-use calibration baseline (a few text-independent offsets per rater) next to any personalised reward model. Beating it on likelihood is necessary but not sufficient evidence of content personalisation; a method that learns content preferences should show it in ranking or choice outcomes, where the offsets cannot help by construction. (iii) Evaluate on raters unseen in training, adapted from k of their own ratings, and on held-out items. (iv) Report calibration (log-likelihood, RPS) and ranking (direction accuracy) separately: they answer different questions — how strongly someone prefers, and what to show them. (v) Check what a learned user representation encodes; in the two published models we reimplemented it was unrelated to scale use ($R^2 \approx 0$) and carried little else that transferred to new raters.

8. Discussion

Why did published personalised models learn little about new raters here? Their evaluations differ from ours in three ways that each favour content-dependent personalisation: synthetic users whose preferences differ in content by construction, binary labels that remove scale use, and users seen during training. On real raters, graded labels and unseen users, the structure that transfers from k ratings to new items is mostly how the rater uses the scale, which neither a content-interacting latent (VPL) nor a mixture of content prototypes (PAL-B) represents ($R^2 \approx 0$ against the offsets).

Adding the three offsets to VPL recovers the whole gain (+0.088), so the two kinds of personalisation are complementary rather than competing; a method that also captures content-dependent individual preferences should be shown to beat VPL plus offsets, not the text-only model. **What this does and does not say about pluralism.** Our results do not show that people agree about content: the item-level disagreement that replicates across rater panels is real (split-half reliability 0.25 on MultiPref) and about a tenth of it is decodable from text. They show that, for predicting an individual new rater from few ratings, that content-level disagreement is a small part of what is predictable, and that demographic or crowd/expert groups add nothing beyond the text. A reward model that serves different groups different content needs evidence that goes beyond likelihood gains on graded ratings — ranking or choice outcomes for those groups — because scale use alone can produce large likelihood gains without changing which response anyone prefers. **Measurement theory as a baseline generator.** Rater severity and scale use are standard facets in many-facet Rasch measurement and in survey methodology. Their transfer to reward modelling costs three parameters per rater and a convex fit, and it should be the first baseline any personalised reward model is compared against.

9. Protocols and what did not hold

All analyses followed timestamped internal protocols (not public registrations); changes after results are logged as amendments with reasons, and the symmetric style model DFs and its hypothesis were added after the first full result. Not supported or not robust: (i) the stage-1 decision rule returned "mixed" (text decodability not ≤ 10 %); (ii) non-inferiority of the 5-parameter DF to the low-rank latent ($\mathrm{DF} - \mathrm{DL} \geq -0.005$) holds with the training-rater prior but not with a few-shot-tuned prior on MultiPref (-0.015) and is formally not met on PRISM (bound -0.006); (iii) "scale beats shift" holds only on MultiPref; (iv) the originally registered probit-family hypothesis is not supported ($\mathrm{F} - \mathrm{D} = -0.029$ [-0.049, -0.008]; superseded when the primary family was moved to the distributional head); (v) the prediction that published user latents encode style failed ($R^2 \approx 0$); (vi) the hybrid's non-inferiority to DFs is formally not met (bound -0.014). Four baseline problems found during the study were fixed in favour of the baselines — a missing constant column in the low-rank latent, a scale non-identifiability in the probit group model, VPL's posterior collapse, and PAL-B's near-unregularised adaptation — and the runs before each fix are reported.

10. Limitations

Frozen-feature text heads for the main tables (a fine-tuned head in §5.4); three datasets, one language; PRISM scores binned to 5 levels; rater offsets are text-independent by design — interactions with content exist in principle and are what the low-rank latent can capture; we test linear decoders for the text-side measurement only; adaptation uses the rater's ratings of other items, not of the same item.

A. Protocols and amendments

All analyses were specified in timestamped internal protocols (PROTOCOL_STAGE1, PROTOCOL_STAGE2, PROTOCOL_STAGE2B, PROTOCOL_STAGE2C); stages 2b and 2c were written after the stage-2a results. Changes after a result were logged as numbered amendments with the reason; the main ones: the first decoding split (a = b = 2) was structurally empty (two ratings determine their distribution) and was replaced by a = 3, b = 1; hand-set penalties in the rater–item model over-fitted and were replaced by inner cross-validation (which reversed an earlier, wrong conclusion that rater shifts do not help); penalty grids were extended when selections fell on grid edges, before any hypothesis was read; the low-rank baseline lacked a constant column and was fixed in its favour; the probit group model had a scale non-identifiability and was fixed; VPL suffered posterior collapse and its KL weight was re-selected including 0; PAL-B's adaptation ridge was re-selected by k-shot validation. Every superseded full run is kept and reported in the supplementary results; two debugging smoke runs and one fold whose output was deleted unread when the penalty grids were extended are not.

B. Hypotheses and outcomes

Stage 2a (MultiPref, k = 20): H1 DF $>$ D — holds (+0.080); H5 as first registered (probit F $>$ D) — not supported (-0.029), superseded by H1 in Amendment 2; H2 DF $>$ group — holds; H3 DF $\geq$ low-rank $-$ 0.005 — holds under the training-rater prior, not robust under a few-shot-tuned prior (-0.015); H4 rater scale $>$ rater shift — holds on MultiPref only; H6 symmetric style model gains likelihood without losing direction accuracy — holds and robust. Replications: H1r, H2r hold on DICES and PRISM; H3r holds on DICES (robust) and is a tie on PRISM under the tuned prior. Stage 2b: B1, B2, B3 hold. Stage 2c: C1 holds (graded), C2 holds (binary: |DFs $-$ D| accuracy -0.0002; all personalisation gains ≤ 0.002 nats), C3 fails (published latents do not encode style, $R^2 \approx 0$), C4 half-holds (+0.088 over VPL; tie with DFs); R1 (PRISM native pairwise choices: every method's own choice-accuracy gain at k = 20 has an upper bound $<$ +0.01) holds (largest bound +0.0071, PAL-B K = 4).

C. Implementation

Text features: last-token final-layer states of Qwen2.5-7B-Instruct under a 5-point judge prompt (MultiPref, PRISM) or a 3-level safety prompt (DICES), label-free PCA to 256 dimensions. Folds: 5 rater folds (stratified by group) $\times$ 5 prompt-group folds; validation: 15 % of training prompt groups. k-shot draws: 5 per k, identical across all models (RC-LoRA: draw 0 only). PRISM item folds are by conversation. Bootstrap: rater clusters, B = 2,000. LoRA: r = 16 on all attention and MLP projections, lr 10^{-4}, one epoch, accumulation 8. VPL / PAL-B: official architectures (latent 512, hidden 512; d_pref 512, mlp2-gelu projectors, angle learner) with an ordered-logistic link for graded labels; VPL's KL weight selected from $\{0.01$ (official), $0.001, 0\}$ and PAL-B's adaptation ridge by k-shot validation (non-official choices made to prevent collapse and over-fitting).

References

Aroyo, L., Taylor, A. S., Díaz, M., Homan, C. M., Parrish, A., Serapio-García, G., Prabhakaran, V., and Wang, D. DICES dataset: Diversity in conversational AI evaluation for safety. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. doi: 10.52202/075280-2321.

Baumgartner, H. and Steenkamp, J.-B. E. M. Response styles in marketing research: A cross-national investigation. *Journal of Marketing Research*, 38(2):143–156, 2001. doi: 10.1509/jmkr.38.2.143.18840.

Bose, A., Xiong, Z., Chi, Y., Du, S. S., Xiao, L., and Fazel, M. LoRe: Personalizing LLMs via low-rank reward modeling, 2025. URL https://arxiv.org/abs/2504.14439.

Chen, D., Chen, Y., Rege, A., Wang, Z., and Vinayak, R. K. PAL: Sample-efficient personalized reward modeling for pluralistic alignment. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

Deng, N., Zhang, X. F., Liu, S., Wu, W., Wang, L., and Mihalcea, R. You are what you annotate: Towards better models through annotator representations. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 12475–12498, 2023.

Dong, Y. R., Hu, T., Liu, Y., Üstün, A., and Collier, N. When personalization meets reality: A multi-faceted analysis of personalized preference learning, 2025. URL https://arxiv.org/abs/2502.19158.

Eckes, T. *Introduction to Many-Facet Rasch Measurement: Analyzing and Evaluating Rater-Mediated Assessments*, volume 22 of *Language Testing and Evaluation*. Peter Lang, Frankfurt am Main, 2011. ISBN 978-3-631-61350-4.

Gordon, M. L., Lam, M. S., Park, J. S., Patel, K., Hancock, J. T., Hashimoto, T., and Bernstein, M. S. Jury learning: Integrating dissenting voices into machine learning models. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, 2022. doi: 10.1145/3491102.3502004.

Kim, G. and Kim, E. Swap-guided preference learning for personalized reinforcement learning from human feedback. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2603.12595.

Kirk, H. R., Whitefield, A., Röttger, P., Bean, A. M., Margatina, K., Ciro, J., Mosquera, R., Bartolo, M., Williams, A., He, H., Vidgen, B., and Hale, S. A. The PRISM alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-3342.

Li, X., Zhou, R., Lipton, Z. C., and Leqi, L. Personalized language modeling from personalized human feedback, 2024.

Linacre, J. M. *Many-Facet Rasch Measurement*. MESA Press, Chicago, 1989.

Miranda, L. J. V., Wang, Y., Elazar, Y., Kumar, S., Pyatkin, V., Brahman, F., Smith, N. A., Hajishirzi, H., and Dasigi, P. Hybrid preferences: Learning to route instances for human vs. AI feedback. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7162–7200, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.355.

Mishra, P., Rastogi, C., Pfohl, S. R., Parrish, A., Teh, T. H., Patel, R., Diaz, M., Wang, D., Paganini, M., Prabhakaran, V., Aroyo, L., and Rieser, V. Decoding safety feedback from diverse raters: A data-driven lens on responsiveness to severity. *Transactions on Machine Learning Research*, 2026. URL https://arxiv.org/abs/2503.05609.

Mostafazadeh Davani, A., Díaz, M., and Prabhakaran, V. Dealing with disagreements: Looking beyond the majority vote in subjective annotations. *Transactions of the Association for Computational Linguistics*, 10:92–110, 2022. doi: 10.1162/tacl_a_00449.

Orlikowski, M., Röttger, P., Cimiano, P., and Hovy, D. The ecological fallacy in annotation: Modelling human label variation goes beyond sociodemographics. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 1017–1029, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-short.88.

Poddar, S., Wan, Y., Ivison, H., Gupta, A., and Jaques, N. Personalizing reinforcement learning from human feedback with variational preference learning. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. doi: 10.52202/079017-1664.

Ryan, M. J., Shaikh, O., Bhagirath, A., Frees, D., Held, W., and Yang, D. SynthesizeMe! inducing persona-guided prompts for personalized reward models in LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. URL https://arxiv.org/abs/2506.05598.

Siththaranjan, A., Laidlaw, C., and Hadfield-Menell, D. Distributional preference learning: Understanding and accounting for hidden context in RLHF. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2312.08358.

Sorensen, T., Moore, J., Fisher, J., Gordon, M. L., Mireshghallah, N., Rytting, C. M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., and Choi, Y. Position: A roadmap to pluralistic alignment. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 46280–46302, 2024.

Zhang, M. J. Q., Wang, Z., Hwang, J. D., Dong, Y., Delalleau, O., Choi, Y., Choi, E., Ren, X., and Pyatkin, V. Diverging preferences: When do annotators disagree and do models know? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 76193–76212, 2025.